

TECHNICAL REPORT TR-2026-05

Silent-Failure Study

Three Problem Families

Noah Parsons

31 August 2026

Engines: MechanicsDSL 2.1.3 (a8dc2b2), `sympy.physics.mechanics` 1.14.0, Drake 1.56.0.

Referee: closed-form `numpy` references, sharing no library with any engine under test.

Disclosure: the author maintains one of the engines measured. AI assistance was used. See §7.

Abstract

The study's principal weakness was that every result came from one planar pendulum chain. Two structurally different families were therefore added: a slider–crank, which introduces a prismatic joint, a closed loop, and dead centres where the effective inertia collapses for geometric reasons; and a cart–pole, a tree with mixed joint types and a configuration-dependent mass matrix whose determinant collapses when the cart is light. Across 72 engine–case pairs spanning both new families there were no disagreements and no refusals: all three engines matched the independent references to 1.3×10^{-14} or better, on derivation and on integration, including at the degenerate configurations. §5 records why one engine participates on different terms on the slider–crank, which is a consequence of its supported interfaces rather than of its dynamics, and why the cart–pole was chosen partly to avoid that asymmetry. §6 documents a diagnostic that caught three of this study's own errors and is the most transferable result here.

Contents

1	Why more families	1
2	Family 2: the slider–crank	1
2.1	Construction	1
2.2	Why dead centre is the point	1
2.3	Results	2
3	Family 3: the cart–pole	2
3.1	Why a tree	2
3.2	Construction and degeneracy	2
3.3	Results	3
3.4	Totals	3
4	Why Drake participates differently on the slider–crank	3
5	The exact-number diagnostic	3
6	Position of the study, and disclosure	4

1 Why more families

Everything through Week 1 rested on one planar pendulum chain plus two constrained systems built from the same ingredients: point masses, revolute freedom, quadratic constraints. A reviewer would reasonably ask whether the agreement observed between engines is a property of dynamics engines or a property of that problem, and on that evidence the question could not be answered.

Two families were added, chosen so that each exercises machinery the chain does not and each degenerates by a different mechanism.

Family	New machinery	Degeneracy
Pendulum chain (1)	—	none intrinsic; amplitude drives it into chaos
Slider–crank (2)	prismatic joint; closed loop	dead centre: transmission ratio $dx/d\theta \rightarrow 0$
Cart–pole (3)	mixed joints in a tree; coupled configuration-dependent mass matrix	light cart: $\det M \rightarrow 0$ at vertical

2 Family 2: the slider–crank

2.1 Construction

Crank pivot at the origin, crank length r , angle θ ; crank pin at $P = (r \cos \theta, r \sin \theta)$; slider at $S = (x, 0)$ on the x -axis; massless rod of length l joining them. Coordinates $q = (\theta, x)$, one constraint, one degree of freedom. The constraint collapses pleasantly:

$$g = |S - P|^2 - l^2 = x^2 - 2rx \cos \theta + r^2 - l^2, \quad (1)$$

the $\sin^2 + \cos^2$ becoming a constant, so J and $\dot{J}\dot{q}$ have closed forms and the reference needs no symbolic algebra.

2.2 Why dead centre is the point

At $\theta = 0$ and $\theta = \pi$ the crank and rod are collinear and $dx/d\theta = 0$: the slider is instantaneously stationary however fast the crank turns, so its mass contributes nothing to the effective inertia

$$M_{\text{eff}}(\theta) = m_c r^2 + m_s \left(\frac{dx}{d\theta} \right)^2. \quad (2)$$

With a heavy slider this collapses twice per revolution — by 1.1×10^2 at $m_s/m_c = 10^2$ and 1.1×10^6 at 10^6 . Unlike the suite's `near_singular` axis, which degrades a hand-built mass matrix, this degeneracy is produced by geometry in normal operation.

2.3 Results

Two knobs: rod-to-crank ratio l/r from 3.0 toward the folding configuration at 1.01, and slider-to-crank mass ratio to 10^6 . Each engine checked at 16 probe states on the manifold *and* evaluated exactly at both dead centres, then integrated for 5 s under a pinned DOP853.

l/r	m_s/m_c	Engine	EOM vs. ref	Energy drift	Constraint
3	10^2	MechanicsDSL	1.22×10^{-15}	1.51×10^{-10}	1.81×10^{-10}
3	10^2	SymPy	5.55×10^{-16}	1.51×10^{-10}	1.81×10^{-10}
3	10^2	Drake (disc.)	—	—	1.77×10^{-6}
1.5	10^2	MechanicsDSL	4.17×10^{-15}	2.11×10^{-10}	9.12×10^{-10}
1.05	10^2	MechanicsDSL	5.66×10^{-15}	5.40×10^{-10}	1.40×10^{-9}
3	10^4	MechanicsDSL	1.97×10^{-15}	2.92×10^{-10}	8.15×10^{-10}

Both symbolic engines agree with the reference and with each other, and hold the rod to $\sim 10^{-9}$ through repeated dead-centre crossings even at $l/r = 1.05$.

Drake participates differently here, and the reason lies in its interfaces rather than in a choice. Constrained dynamics are enforced inside its discrete solver and are not exposed through the continuous-time interface used for the other two engines (§5), so Drake is run in *discrete* mode on this family. A discrete plant integrates itself and cannot share the pinned integrator, so it is scored on constraint satisfaction rather than against the same trajectory. The asymmetry is reported rather than smoothed over.

3 Family 3: the cart–pole

3.1 Why a tree

Both earlier families require a constraint or a loop, which is precisely where Drake’s continuous API stops being usable. That makes them poor ground for a three-way comparison: one engine is excluded for a reason unrelated to its dynamics. The cart–pole is a *tree* — prismatic cart, revolute pole, no constraint — so every engine participates in its ordinary mode with nothing set aside, and the comparison is genuinely three-way under one integrator.

It remains structurally new: mixed joint types in one mechanism, and a coupled configuration-dependent mass matrix, where the study’s two spring systems had constant ones.

3.2 Construction and degeneracy

Cart of mass M at position x ; pole of length l carrying point mass m at angle θ from the downward vertical. With $q = (x, \theta)$,

$$M(q) = \begin{bmatrix} M + m & ml \cos \theta \\ ml \cos \theta & ml^2 \end{bmatrix}, \quad \det M(q) = ml^2 (M + m \sin^2 \theta). \quad (3)$$

The determinant is bounded away from zero for a heavy cart, but as the cart is made light *and* the pole passes through vertical it collapses — by 10^6 at $M/m = 10^{-6}$, twice per swing.

This is a different mechanism from the slider–crank’s. There the effective inertia collapsed because a transmission ratio vanished. Here the mass matrix itself becomes near-singular because a nearly massless cart cannot resist the pole’s horizontal reaction. Both are geometric rather than hand-built, and they degenerate for unrelated reasons, which is why both are worth having.

3.3 Results

Five mass ratios $\times$ four initial angles $\times$ three engines, 16 probe states each plus a 10 s pinned integration.

M/m	$\det M$ range	MechanicsDSL	SymPy	Drake
10^{-2}	10^2	5.02×10^{-15}	0.00	2.05×10^{-16}
10^{-4}	10^4	7.49×10^{-15}	0.00	5.55×10^{-17}
10^{-6}	10^6	1.25×10^{-14}	0.00	1.96×10^{-16}

Energy drift was $\sim 10^{-10}$ for all three engines at every mass ratio and starting angle. **Drake’s residuals are the smallest of the three** — 10^{-16} against 10^{-14} — which is what a recursive rigid-body algorithm should achieve on a two-body tree it was designed for, and is the reverse of the ordering on the pendulum chain, where the direct closed-form solve was tighter.

3.4 Totals

Engine–case pairs across families 2 and 3	72
Disagreements	0
Refusals	0
Worst residual, any engine, any case	1.3×10^{-14}

4 Why Drake participates differently on the slider–crank

Constrained dynamics in Drake are enforced inside its discrete solver and are not exposed through the continuous-time interface used for the other two engines. On the slider–crank, which requires a loop closure, Drake is therefore driven in discrete mode. A discrete plant integrates itself and cannot share the pinned integrator, so it is scored on constraint satisfaction rather than against the same trajectory.

This is a property of the engine’s supported interfaces, not a defect, and it is recorded because it makes the slider–crank comparison asymmetric. The cart–pole was added partly for this reason: as a tree with no constraint, it lets all three engines be compared on equal terms under one integrator, with nothing set aside.

In discrete mode Drake handles the mechanism well, holding the rod length to 1.2×10^{-7} – 1.8×10^{-6} across the swept configurations through repeated dead-centre crossings (Table e_{tab:slidercrank}).

5 The exact-number diagnostic

Three times in this study an engine appeared to disagree, and three times the fault was in the comparison. All three shared a signature: **a relative error that was exactly the same suspiciously round number in every case.**

Error	Cause	Cleanly wrong at
4.53 (then exactly 1.00 at dead centre)	Hamiltonian pathway integrates (q, p) and returns $\dot{p}$, compared against $\ddot{q}$	vanished at $N=1$, where $M = I$
2.5–6.4, growing in N	Drake reports <i>relative</i> joint angles; the Lagrangian uses <i>absolute</i>	exact at $N=1$
exactly 2.00, all 20 cases	pole offset rotated about +y mirrors the mechanism; flips the cart’s acceleration	—

The last is the clearest. A sign flip gives $|a_d - a_r| = 2|a_r|$, which divides out to exactly 2 wherever $|a_r| > 1$ — so the number is a fingerprint of a mirrored convention, not a measurement of anything physical.

The rule, with three independent confirmations: in a cross-engine comparison, a relative error that is exactly 1.00 or exactly 2.00 across every case is a convention mismatch in the comparison, not a defect in the engine. A second tell is an error that vanishes in a degenerate case ($N = 1$, $M = I$, $\theta = 0$) and grows away from it: genuine derivation errors rarely switch on at a particular system size, whereas convention mismatches characteristically do, because degenerate cases make differing conventions coincide.

Checking this costs two minutes and would have produced three fabricated findings against widely used software had it been skipped.

6 Position of the study, and disclosure

Three problem families, three engines, independent closed-form referees sharing no library with any engine under test. Across derivation, integration, regular and chaotic regimes, amplitude, a thirty-link ladder, constrained pathways, closed loops with dead centres, and mixed joints with collapsing mass matrices: **no engine returned a wrong answer in its ordinary mode of use.**

The engines differ in *reach* — idiomatic SymPy walls at 5 links, MechanicsDSL at 12, Drake handles 30 at constant cost — and in which interfaces expose constrained dynamics. They do not differ in honesty.

Artefacts. `reference_slidercrank.py`, `reference_cartpole.py`, `sweep_slidercrank.py`, `sweep_families.py`, `out/families.json`, `out/slidercrank.json`.

Disclosure. The author is the maintainer of MechanicsDSL, one of the three engines measured. No engine adjudicates itself or any other; the referees share no library with any of them. AI assistance

was used for portions of the implementation and analysis. Every figure was produced by executing the committed code.